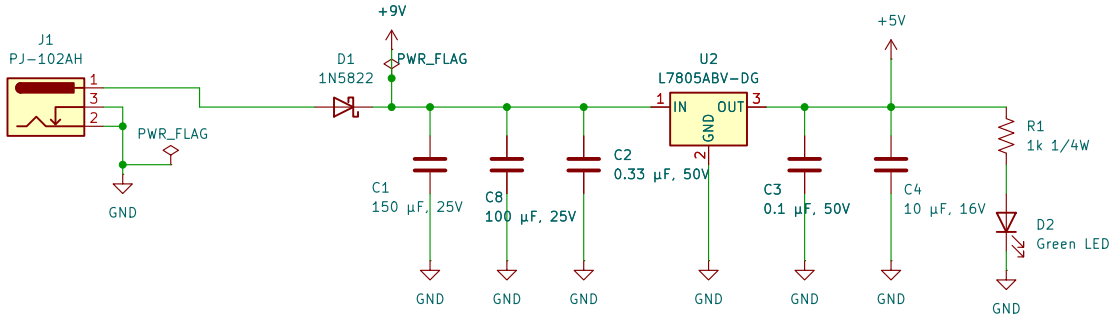
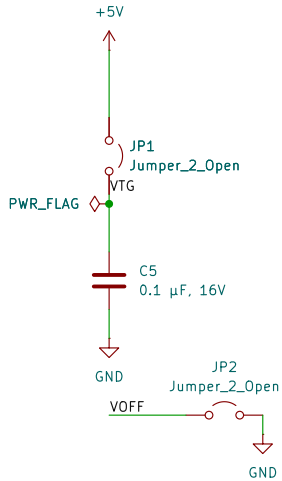
POWER: 9V input, reverse-polarity diode, 5V linear regulator



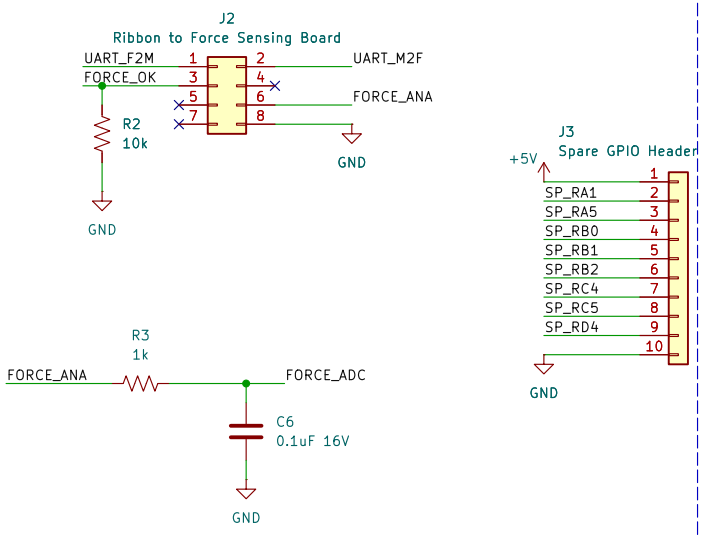
D1 provides reverse-polarity protection. C1 and C8 provide bulk capacitance for supply filtering and motor current transients. C2 provides high-frequency decoupling. The DRV8871 provides motor current regulation. The L7805 requires appropriate thermal management.

NANO POWER: isolation jumpers and VTG bypass



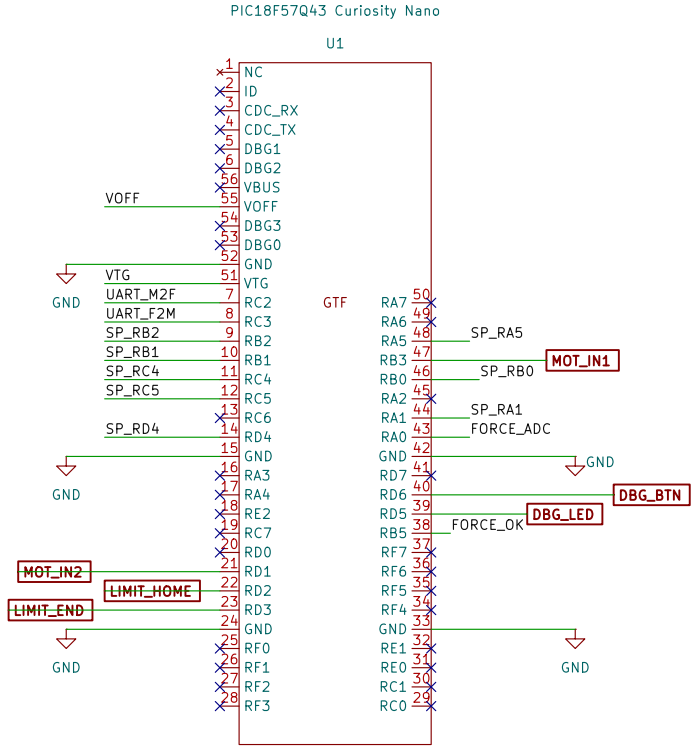
With the 9V adapter: JP1 and JP2 fitted. USB-only operation: remove both jumpers. JP1 connects regulated +5V to VTG, and JP2 connects VOFF to GND. C5 provides 0.1 µF bypass capacitance at VTG.

RIBBON CONNECTOR, FORCE_ANA FILTER AND SPARE GPIO HEADER



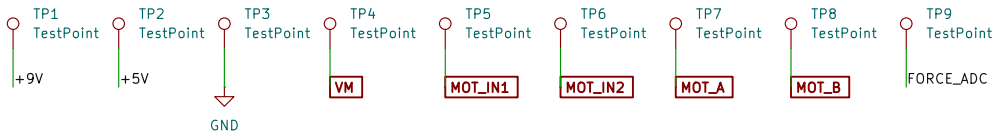
J2 matches the Force Sensing Board ribbon connector. Pin 8 = GND. R2 holds FORCE_OK LOW when the cable is unplugged, so the motor stays off. R3/C6 filter FORCE_ANA before the ADC (RA0)

MICROCONTROLLER: PIC18F57Q43 Curiosity Nano



PIC18F57Q43 Curiosity Nano controls the motor through RB3 (PWM) and RD1 (direction). RD2/RD3 monitor limit switches. RC2/RC3 provide UART communication. RB5 receives FORCE_OK, and RA0 reads FORCE_ADC. RD5/RD6 support debugging. VTG/VOFF manage Nano power, and unused GPIO pins connect to J3.

TEST POINTS



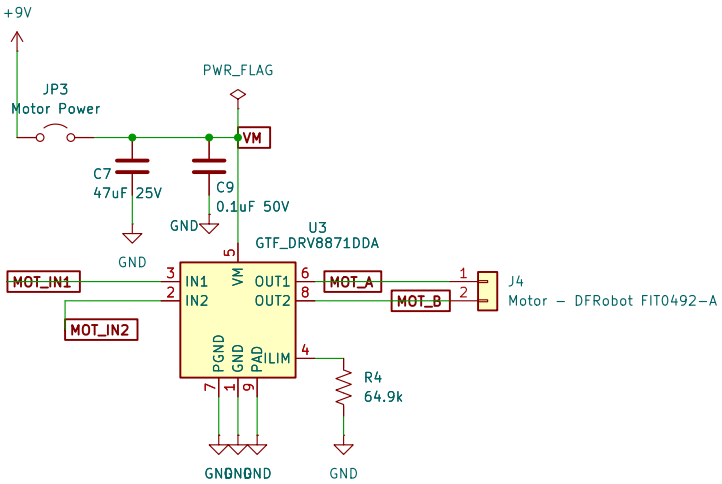
Gabriel Toneser Facchin
Team 103 - GripRx

Sheet: /
File: GripRx_Motor_Board.kicad_sch

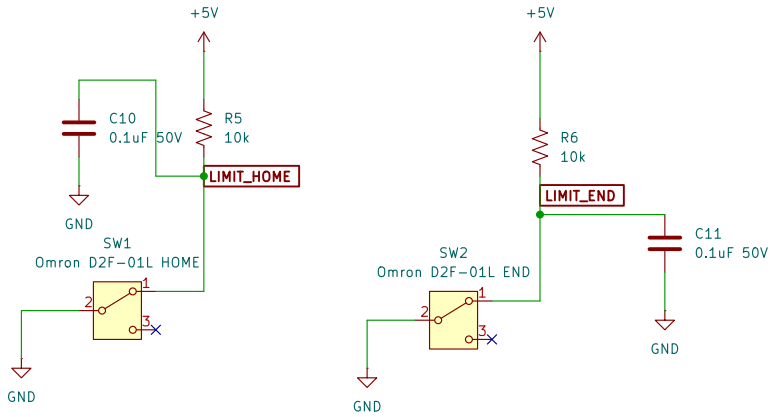
Title: GripRx Motor Board

Size: A4	Date: 2026-10-25	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/2

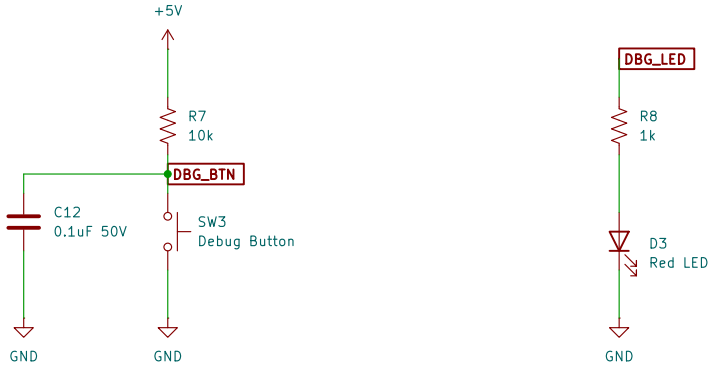
ACTUATOR WITH DRIVER
DRV8871 H-bridge and DC gear motor



LIMIT SWITCHES
Home and end switches



DEBUG - Button (RD6) and LED (RD5)



Gabriel Toneser Facchin
Team 103 - GripRx

Sheet: /Motor Driver and Limits/
File: Motor_Driver_and_Limits.kicad_sch

Title: GripRx Motor Board - Motor Driver

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/2